

ON THE DE RHAM COHOMOLOGY OF CYCLIC COVERS

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ABSTRACT. We compute explicit bases for the de Rham cohomology of cyclic covers of the projective line defined over an algebraically closed field of characteristic $p \geq 0$. For both Kummer and Artin-Schreier extensions, we describe precise k -bases for the cohomology groups $H^1(X, \mathcal{O}_X)$ and $H^0(X, \Omega_X)$, and we use these to construct an explicit basis for the first de Rham cohomology group $H_{\text{dR}}^1(X/k)$ via ech cohomology. Our approach relies on detailed computations of divisors of functions and differentials, together with residue calculations and the duality pairing between $H^0(X, \Omega_X)$ and $H^1(X, \mathcal{O}_X)$. The resulting expressions are given in closed form in terms of the defining equation of the cover, making the cohomology fully explicit and readily applicable to questions involving group actions, Hodge-de Rham filtrations, and the study of p -cyclic covers.

1. INTRODUCTION

There are few classes of curves where explicit computations can be made. Cyclic covers of curves which generalize hyperelliptic curves are among them. H. Hasse in [7] provided a unified treatment of both Artin-Schreier and Kummer extensions of the rational function fields and H. Bocklandt [1] gave explicit bases of holomorphic differentials on them. Furthermore, the work of Sekiguchi, Oort, and Suwa [12], [13], [14] highlights deeper connections between Kummer and Artin-Schreier-type extensions via group scheme theory, suggesting that a unified framework for these computations is possible, [2], [8].

On the other hand there is a lot of interest concerning study of the de Rham cohomology of algebraic curves over fields of arbitrary characteristic especially with respect to their Galois module structure and the splitting problem of the exact sequence

$$1 \rightarrow H^0(X, \Omega_X) \rightarrow H_{\text{dR}}^1(X) \rightarrow H^1(X, \mathcal{O}_X) \rightarrow 0,$$

seen as Galois module, see [3], [4], [5], [6], [9]. For cyclic covers of the projective line, the geometry and arithmetic of the curve are encoded in an explicit equation, and this allows one to perform cohomological computations in a concrete manner. Nevertheless, even in this relatively accessible class of curves, an explicit description of the Hodge de Rham exact sequence and the associated bases for $H_{\text{dR}}^1(X/k)$ remains delicate.

The aim of this article is to give explicit k -bases for the cohomology groups $H^1(X, \mathcal{O}_X)$, $H^0(X, \Omega_X)$, and the first de Rham cohomology group $H_{\text{dR}}^1(X/k)$ when X is a cyclic cover of the projective line of either Kummer or Artin-Schreier type. Our approach relies on ech cohomology with respect to the natural affine covering

Date: November 24, 2025.

2020 Mathematics Subject Classification. Primary 14F40; Secondary 14H30, 14H37.

Key words and phrases. De Rham Cohomology, Artin-Schreier and Kummer Covers.

induced by the projection $X \rightarrow \mathbb{P}^1$, combined with explicit computations of divisors of functions and differentials. This method allows us to describe bases in completely elementary terms, making the cohomology transparent in examples and suitable for applications to deformation theory, G -actions on cohomology, and the study of the Hodge Rham sequence.

The structure of the paper is as follows. In Section 2 we treat the case of Kummer extensions, beginning with divisor computations, then exhibiting bases for $H^1(X, \mathcal{O}_X)$ and $H^0(X, \Omega_X)$, and finally constructing an explicit basis for $H_{\text{dR}}^1(X/k)$. Section 3 contains the analogous results for ArtinSchreier extensions, where the arithmetic of characteristic p introduces new phenomena. Throughout both sections, duality pairings and residue computations play a central role.

Acknowledgements The research project is implemented in the framework of H.F.R.I Call Basic research Financing (Horizontal support of all Sciences) under the National Recovery and Resilience Plan Greece 2.0 funded by the European Union Next Generation EU(H.F.R.I. Project Number: 14907).



2. KUMMER EXTENSIONS

2.1. Computations. We consider the curve given as a Kummer extension of $k(x)$, with k an algebraically closed field of characteristic $p > 0$ given by an equation of the form

$$X : y^n = f(x) = \prod_{i=1}^r (x - \rho_i)^{l_i}.$$

We recall some facts about these curves, see [1], [10], [11]. Without loss of generality we can assume that $l := \deg f(x) \equiv 0 \pmod{n}$. Thus we get that the points above ∞ are not ramified in the cover $X \rightarrow \mathbb{P}^1$. The ρ_i 's are the only points of $\mathbb{P}^1$ in the branch locus of the cover, and each point above $x = \rho_i$ is ramified with ramification index $e_i = \frac{n}{(n, l_i)}$. The points of X lying above each ρ_i are $P_{i,1}, \dots, P_{i,g_i}$, where $g_i = (n, l_i)$, while the points above ∞ are $P_{\infty,1}, \dots, P_{\infty,n}$.

If $0 \in \mathbb{P}^1$ is in the branch locus, then its corresponding exponent in the equation of X will also be denoted as l_0 , the number of points above it equals $g_0 = (n, l_0)$, and their ramification index is $e_0 = \frac{n}{(n, l_0)}$. If 0 is not in the branch locus, then $l_0 := n, g_0 := n$, and $e_0 := 1$.

We compute

$$\begin{aligned} (x) &= e_0 \sum_{j=1}^{g_0} P_{0,j} - \sum_{m=1}^n P_{\infty,m} \\ (y) &= \sum_{i=1}^r \lambda_i \sum_{j=1}^{g_i} P_{i,j} - t \sum_{m=1}^n P_{\infty,m} \end{aligned}$$

where

$$t = l/n, \text{ and } \lambda_i = \frac{e_i l_i}{n} = \frac{l_i}{(l_i, n)}.$$

For each μ , $\mu \in \{1, 2, \dots, n-1\}$, we define $m_i^{(\mu)}$, $v_i^{(\mu)}$ by

$$\mu \lambda_i = \mu \frac{l_i}{(l_i, n)} = m_i^{(\mu)} e_i + v_i^{(\mu)}, \text{ with } 0 \leq v_i^{(\mu)} \leq e_i - 1.$$

We set

$$(1) \quad g_\mu(x) = \prod_{i=1}^r (x - \rho_i)^{m_i^{(\mu)}} \text{ for } \mu \in \{1, 2, \dots, n-1\}$$

and we compute

$$(2) \quad \left(\frac{y^\mu}{g_\mu(x)} \right) = \sum_{i=1}^r v_i^{(\mu)} \sum_{j=1}^{g_i} P_{i,j} - t^{(\mu)} \sum_{m=1}^n P_{\infty,m},$$

where

$$t^{(\mu)} = \frac{1}{n} \sum_{i=1}^r g_i v_i^{(\mu)}.$$

2.2. A k -vector space bases for $H^1(X, \mathcal{O}_X)$, $H^0(X, \Omega_X)$. We begin by computing a basis for $H^1(X, \mathcal{O}_X)$. We denote by $\pi : X \rightarrow \mathbb{P}^1$ a covering map from X to $\mathbb{P}^1$. For every point $a \in \mathbb{P}^1$ we define $U_a := X \setminus \pi^{-1}(a)$. Observe that $x^i y^j$ for $i, j \geq 0$ are elements of $\mathcal{O}_X(U_\infty)$. Also by equation (2) the elements $\frac{y^\mu}{g_\mu(x)}$ are in $\mathcal{O}_X(U_\infty)$, while the elements

$$\frac{1}{x^\nu} \frac{y^\mu}{g_\mu(x)} \text{ for } \nu \geq t^{(\mu)} \text{ and } \frac{1}{x^\nu} \text{ for } \nu \geq 0,$$

are in $\mathcal{O}_X(U_0)$. We identify the first sheaf cohomology group of $\mathcal{O}_X$ with the corresponding first ech cohomology group, so

$$H^1(X, \mathcal{O}_X) = \frac{\mathcal{O}_X(U_0 \cap U_\infty)}{\{f_0 - f_\infty : f_0 \in \mathcal{O}_X(U_0), f_\infty \in \mathcal{O}_X(U_\infty)\}}.$$

The k -vector space $H^1(X, \mathcal{O}_X)$ admits as basis the following linear equivalence classes

$$h_{\mu,\nu} := \left[\frac{1}{x^\nu} \frac{y^\mu}{g_\mu(x)} \right], 1 \leq \mu \leq n-1 \text{ s.t. } t^{(\mu)} \geq 2, 1 \leq \nu \leq t^{(\mu)} - 1.$$

The number of these classes equals

$$\sum_{\mu=1}^{n-1} (t^{(\mu)} - 1),$$

which are equal to the genus of the curve X according to [1, eq. (47)].

We will now give a basis for the space $H^0(X, \Omega_X)$. By [1, Satz 16] we get that a basis of holomorphic differentials is given by

$$(3) \quad \omega_{\mu,\nu} := x^{\nu-1} \frac{g_\mu(x)}{y^\mu} dx, \mu \in \{1, \dots, n-1\} \text{ s.t. } t^{(\mu)} \geq 2, 1 \leq \nu \leq t^{(\mu)} - 1.$$

For our description of $H^1(X, \mathcal{O}_X)$ using ech cohomology, we can see from Tait's dissertation [16, sec. 2.5] that Serre's duality map becomes (the coefficient is not

essential to the definition of the map, but will be useful later)

$$\begin{aligned} \langle \cdot, \cdot \rangle : H^0(X, \Omega_X) \times H^1(X, \mathcal{O}_X) &\longrightarrow k \\ \langle \omega, [f] \rangle &\longmapsto \left(-\frac{1}{n} \right) \sum_{P \in \pi^{-1}(\infty)} \text{Res}_P(f\omega), \end{aligned}$$

where $\text{Res}_P(\omega)$ is the residue of the differential ω at the point $P \in X$ [15, Chap. II, 7,8,12]. We want to show that $\{h_{\mu,\nu}\}_{\mu,\nu}$ is the dual basis to $\{\omega_{\mu,\nu}\}_{\mu,\nu}$, i.e.

$$\langle \omega_{\mu_1,\nu_1}, h_{\mu_2,\nu_2} \rangle = \delta_{\mu_1,\mu_2} \delta_{\nu_1,\nu_2}.$$

We will first recall the definition of the residue of a differential. Let F be the function field of X , denote by $\widehat{F}_P = k((t))$ the completion of F for the valuation v_P , where t is a local uniformizer at P . We identify $f \in F$ with its image in $\widehat{F}_P$ which is called the Laurent expansion of f at P :

$$f = \sum_{i > -\infty} c_i t^i, c_i \in k.$$

The residue $\text{Res}_P(\omega)$ is defined as the coefficient c_{-1} of the Laurent expansion of f in a decomposition $\omega = f dt$.

Some important properties for the residue can be found in no.11 of [15]:

- (1) $\text{Res}_P(\omega)$ is k -linear in ω .
- (2) $\text{Res}_P(\omega) = 0$ if $v_P(\omega) \geq 0$.
- (3) $\text{Res}_P(dg) = 0$ for all $g \in \widehat{F}_P$.
- (4) $\text{Res}_P(dg/g) = v_P(g)$ for all $g \in \widehat{F}_P \setminus \{0\}$.

We will also need the following lemma [15, lemma 4 no. 12]:

Lemma 1. *If we have a covering map $\varphi : X \rightarrow \mathbb{P}^1$ of curves, then for every point $P \in \mathbb{P}^1$,*

$$\sum_{Q \rightarrow P} \text{Res}_Q(\omega) = \text{Res}_P(\text{Tr}(\omega)),$$

the sum being over all the points $Q \in X$ such that $\varphi(Q) = P$.

In our case we have the covering map

$$\begin{aligned} \pi : X &\rightarrow \mathbb{P}^1 \\ (x, y) &\mapsto x, \end{aligned}$$

which corresponds to the Galois extension of function fields F/E , where $E = k(x)$ is the function field of $\mathbb{P}^1$. If $\omega = f dx$, then

$$\text{Tr}(\omega) = \text{Tr}_{F/E}(\omega) = \text{Tr}_{F/E}(f) dx.$$

Therefore we get

$$\sum_{m=1}^n \text{Res}_{P_{\infty,m}}(\omega) = \text{Res}_{\infty}(\text{Tr}(\omega)) = \text{Res}_{\infty}(\text{Tr}_{F/E}(f) dx).$$

Proposition 2. *The bases given above for $H^0(X, \Omega_X)$ and $H^1(X, \mathcal{O}_X)$ are dual to each other.*

Proof. We calculate

$$\begin{aligned}\langle \omega_{\mu_1, \nu_1}, h_{\mu_2, \nu_2} \rangle &= -\frac{1}{n} \sum_{P \rightarrow \infty} \text{Res}_P \left(\frac{x^{\nu_1 - \nu_2} y^{\mu_2 - \mu_1} g_{\mu_1}(x) dx}{g_{\mu_2}(x)} \frac{dx}{x} \right) \\ &= -\frac{1}{n} \text{Res}_\infty \left(\text{Tr}_{F/E} \left(\frac{x^{\nu_1 - \nu_2} y^{\mu_2 - \mu_1} g_{\mu_1}(x) dx}{g_{\mu_2}(x)} \frac{dx}{x} \right) \right).\end{aligned}$$

If $\mu_1 = \mu_2, \nu_1 = \nu_2$, then

$$\langle \omega_{\mu_1, \nu_1}, h_{\mu_2, \nu_2} \rangle = -\frac{1}{n} \text{Res}_\infty \left(n \frac{dx}{x} \right) = \text{Res}_\infty \left(\frac{d\left(\frac{1}{x}\right)}{\frac{1}{x}} \right) = 1,$$

using the properties of the residue mentioned above.

Suppose now that either $\mu_1 \neq \mu_2$ or $\nu_1 \neq \nu_2$. Since the extension is Galois and only y is not in the base field of the extension, we have

$$\text{Tr}_{F/E} \left(\frac{x^{\nu_1 - \nu_2 - 1} y^{\mu_2 - \mu_1} g_{\mu_1}(x)}{g_{\mu_2}(x)} \right) = \frac{x^{\nu_1 - \nu_2 - 1} g_{\mu_1}(x)}{g_{\mu_2}(x)} \text{Tr}_{F/E}(y^{\mu_2 - \mu_1}).$$

Since we are in the Kummer extension case, $\text{Gal}(E/F) = \langle \sigma \rangle$ where

$$\begin{aligned}\sigma : F &\rightarrow F \\ x &\mapsto x \\ y &\mapsto y \cdot \zeta_n\end{aligned}$$

where ζ_n is a primitive n -th root of unity. Thus

$$(4) \quad \text{Tr}_{F/E}(y^{\mu_2 - \mu_1}) = \sum_{i=0}^{n-1} \sigma^i(y^{\mu_2 - \mu_1}) = y^{\mu_2 - \mu_1} \sum_{i=0}^{n-1} \zeta_n^{i(\mu_2 - \mu_1)}.$$

If $\mu_1 = \mu_2$ and $\nu_1 \neq \nu_2$, then

$$\langle \omega_{\mu_1, \nu_1}, h_{\mu_2, \nu_2} \rangle = -\frac{1}{n} \text{Res}_\infty (n x^{\nu_1 - \nu_2 - 1} dx).$$

If $\nu_1 > \nu_2$, then $\nu_1 - \nu_2 - 1 \geq 0$ and

$$\begin{aligned}\text{Res}_\infty (n x^{\nu_1 - \nu_2 - 1} dx) &= \text{Res}_\infty \left(-n \frac{1}{x^{-(\nu_1 - \nu_2 - 1)}} x^2 d\left(\frac{1}{x}\right) \right) \\ &= \text{Res}_\infty \left(-n \frac{1}{x^{-(\nu_1 - \nu_2 + 1)}} d\left(\frac{1}{x}\right) \right)\end{aligned}$$

since $d\left(\frac{1}{x}\right) = -\frac{1}{x^2} dx$. Since $\nu_1 - \nu_2 + 1 \geq 2$, from the definition of the residue we see that

$$\text{Res}_\infty \left(-n \frac{1}{x^{-(\nu_1 - \nu_2 + 1)}} d\left(\frac{1}{x}\right) \right) = 0.$$

If $\nu_1 < \nu_2$, then working similarly we see that again $\langle \omega_{\mu_1, \nu_1}, h_{\mu_2, \nu_2} \rangle = 0$.

If $\mu_1 \neq \mu_2$, then the sum of roots of unity appearing in eq. (4) is known to be zero.

In conclusion,

$$\langle \omega_{\mu_1, \nu_1}, h_{\mu_2, \nu_2} \rangle = 0$$

when either $\mu_1 \neq \mu_2$ or $\nu_1 \neq \nu_2$. □

2.3. A k -vector space basis for $H_{dR}^1(X/k)$. Using the same description for the first de Rham cohomology group of X (via ech cohomology) as the one given in [9] we have that $H_{dR}^1(X/k)$ can be calculated by the quotient

$$\frac{\{(\omega_0, \omega_\infty, f_{0\infty}) \in \Omega_X(U_0) \times \Omega_X(U_\infty) \times \mathcal{O}_X(U_0 \cap U_\infty) \mid df_{0\infty} = \omega_0|_{U_0 \cap U_\infty} - \omega_\infty|_{U_0 \cap U_\infty}\}}{\{(df_0, df_\infty, f_0|_{U_0 \cap U_\infty} - f_\infty|_{U_0 \cap U_\infty}) \mid f_0 \in \mathcal{O}_X(U_0), f_\infty \in \mathcal{O}_X(U_\infty)\}}.$$

There are also canonical maps

$$\begin{aligned} i : H^0(X, \Omega_X) &\rightarrow H_{dR}^1(X/k) \\ \omega &\mapsto [(\omega|_{U_0}, \omega|_{U_\infty}, 0)] \end{aligned}$$

and

$$\begin{aligned} p : H_{dR}^1(X/k) &\rightarrow H^1(X, \mathcal{O}_X) \\ [(\omega_0, \omega_\infty, f_{0\infty})] &\mapsto [f_{0\infty}]. \end{aligned}$$

From Proposition 4.1.2 of Tait's dissertation [16] we get that the following short sequence is exact (the result is about hyperelliptic curves, but it can be extended with no changes in the proof in our setting):

$$0 \rightarrow H^0(X, \Omega_X) \xrightarrow{i} H_{dR}^1(X/k) \xrightarrow{p} H^1(X, \mathcal{O}_X) \rightarrow 0,$$

hence the basis of the k -vector space $H_{dR}^1(X/k)$ can be calculated from the bases of $H^0(X, \Omega_X)$ and $H^1(X, \mathcal{O}_X)$.

In order to write down an explicit k -basis for the first de Rham cohomology group, we will need the following notation: Say we have $h(x) \in k[x]$ and $m \in \mathbb{N}$, then $h^{\leq m}(x)$ (respectively $h^{> m}(x)$) is the sum of monomials of $h(x)$ with degree $\leq m$ (respectively $> m$).

Theorem 3. *Let $I^{(\mu)} \subset \mathbb{N}$ defined by*

$$I^{(\mu)} := \{1 \leq i \leq r \mid v_i^{(\mu)} \neq 0\}.$$

Set

$$\psi_{\mu, \nu}(x) := \sum_{i=1}^r \left(g_i v_i^{(\mu)} x \prod_{j \in I^{(\mu)} \setminus \{i\}} (x - \rho_j) \right) - \nu n \prod_{i \in I^{(\mu)}} (x - \rho_i) \in k[x].$$

Then the elements

$$a_{\mu, \nu} := \left[\left(\frac{\psi_{\mu, \nu}^{\leq \nu+1}(x)}{n x^{\nu+1}} \frac{g_{n-\mu}(x)}{y^{n-\mu}} dx, \frac{\psi_{\mu, \nu}^{> \nu+1}(x)}{n x^{\nu+1}} \frac{g_{n-\mu}(x)}{y^{n-\mu}} dx, \frac{y^\mu}{x^\nu g_\mu(x)} \right) \right]$$

$$\text{for } 1 \leq \mu \leq n-1 \text{ s.t. } t^{(\mu)} \geq 2 \text{ and } 1 \leq \nu \leq t^{(\mu)} - 1$$

together with the elements

$$\delta_{\mu, \nu} := \left[\left(\frac{x^{\nu-1} g_\mu(x)}{y^\mu} dx, \frac{x^{\nu-1} g_\mu(x)}{y^\mu} dx, 0 \right) \right]$$

$$\text{for } 1 \leq \mu \leq n-1 \text{ s.t. } t^{(\mu)} \geq 2 \text{ and } 1 \leq \nu \leq t^{(\mu)} - 1$$

form a basis for $H_{dR}^1(X/k)$ as a k -vector space.

Proof. It can be easily seen that the projections of the $a_{\mu, \nu}$'s form the basis of $H^1(X, \mathcal{O}_X)$ and that the unique preimages of the $\delta_{\mu, \nu}$'s form the basis of $H^0(X, \Omega_X)$. Therefore, if the $a_{\mu, \nu}$'s and $\delta_{\mu, \nu}$'s are well-defined elements of $H_{dR}^1(X/k)$, then they indeed form a k -basis for this space. The $\delta_{\mu, \nu}$'s are evidently well-defined, so we focus on the $a_{\mu, \nu}$'s.

First, we show that $\phi_\mu(x) := \left(\frac{y^\mu}{g_\mu(x)}\right)^n$ is a polynomial of x :

$$\begin{aligned}\phi_\mu(x) &= \left(\frac{y^\mu}{g_\mu(x)}\right)^n = \frac{f(x)^\mu}{g_\mu(x)^n} = \frac{\prod_{i=1}^r (x - \rho_i)^{\mu l_i}}{\prod_{i=1}^r (x - \rho_i)^{n m_i^{(\mu)}}} \\ &= \prod_{i=1}^r (x - \rho_i)^{v_i^{(\mu)}(n, l_i)}.\end{aligned}$$

Because we'll need it later we write here a convenient expression for the formal derivative of $\log(\phi_\mu(x))$:

$$\begin{aligned}(5) \quad \phi'_\mu(x) &= \left(\prod_{i=1}^r (x - \rho_i)^{v_i^{(\mu)}(n, l_i)}\right)' = \phi_\mu(x) \sum_{i=1}^r \frac{v_i^{(\mu)}(n, l_i)}{x - \rho_i} \\ &\Rightarrow \frac{\phi'_\mu(x)}{\phi_\mu(x)} = \sum_{i=1}^r \frac{v_i^{(\mu)}(n, l_i)}{x - \rho_i} = \sum_{i \in I^{(\mu)}} \frac{v_i^{(\mu)}(n, l_i)}{x - \rho_i}.\end{aligned}$$

Now, since we cannot do much to calculate the differential of $\frac{y^\mu}{x^\nu g_\mu(x)}$, we'll calculate the differential of its n -th power instead:

$$\begin{aligned}d\left(\left(\frac{y^\mu}{x^\nu g_\mu(x)}\right)^n\right) &= d\left(\frac{\phi_\mu(x)}{x^{\nu n}}\right) \\ &= \frac{\phi'_\mu(x)x^{\nu n} - \phi_\mu(x)\nu n x^{\nu n-1}}{x^{2\nu n}} dx \\ &= \frac{\phi'_\mu(x)}{x^{\nu n}} dx - \frac{\phi_\mu(x)\nu n}{x^{\nu n+1}} dx.\end{aligned}$$

Also

$$d\left(\left(\frac{y^\mu}{x^\nu g_\mu(x)}\right)^n\right) = n \left(\frac{y^\mu}{x^\nu g_\mu(x)}\right)^{n-1} d\left(\frac{y^\mu}{x^\nu g_\mu(x)}\right)$$

hence

$$n \left(\frac{y^\mu}{x^\nu g_\mu(x)}\right)^{n-1} d\left(\frac{y^\mu}{x^\nu g_\mu(x)}\right) = \frac{\phi'_\mu(x)}{x^{\nu n}} dx - \frac{\phi_\mu(x)\nu n}{x^{\nu n+1}} dx$$

which implies

$$\begin{aligned}(6) \quad d\left(\frac{y^\mu}{x^\nu g_\mu(x)}\right) &= \frac{1}{n} \left(\frac{x^\nu g_\mu(x)}{y^\mu}\right)^{n-1} \left(\frac{\phi'_\mu(x)}{x^{\nu n}} dx - \frac{\phi_\mu(x)\nu n}{x^{\nu n+1}} dx\right) \\ &= \left(\frac{y^\mu}{g_\mu(x)}\right) \left(\frac{1}{n x^{\nu+1}}\right) \left(\frac{x \phi'_\mu(x)}{\phi_\mu(x)} - \nu n\right) dx \\ &= \left(\frac{y^\mu \cdot y^{n-\mu}}{g_\mu(x) \cdot g_{n-\mu}(x)}\right) \left(\frac{1}{n x^{\nu+1}}\right) \left(\frac{x \phi'_\mu(x)}{\phi_\mu(x)} - \nu n\right) \left(\frac{g_{n-\mu}(x)}{y^{n-\mu}} dx\right) \\ &= \frac{1}{n x^{\nu+1}} \left(\frac{f(x)}{g_\mu(x) \cdot g_{n-\mu}(x)}\right) \left(\frac{x \phi'_\mu(x)}{\phi_\mu(x)} - \nu n\right) \omega_{n-\mu, 1},\end{aligned}$$

where $\omega_{n-\mu, 1} \in H^0(X, \Omega_X)$ is a holomorphic differential as defined in eq. (3).

From the definition of $g_\mu(x)$ in eq. (1), we know that:

$$g_\mu(x) \cdot g_{n-\mu}(x) = \prod_{i=1}^r (x - \rho_i)^{m_i^{(\mu)} + m_i^{(n-\mu)}}.$$

For the exponents appearing in the above expression we have that for $i \in \{1, \dots, r\}$ and $\mu \in \{1, \dots, n-1\}$:

$$\begin{aligned}\mu\lambda_i &= m_i^{(\mu)}e_i + v_i^{(\mu)} \\ (n-\mu)\lambda_i &= m_i^{(n-\mu)}e_i + v_i^{(n-\mu)}\end{aligned}$$

and adding these two equations we get (remember $\lambda_i = \frac{l_i}{(n, l_i)}$ and $e_i = \frac{n}{(n, l_i)}$):

$$n \frac{l_i}{(n, l_i)} = (m_i^{(\mu)} + m_i^{(n-\mu)}) \frac{n}{(n, l_i)} + (v_i^{(\mu)} + v_i^{(n-\mu)}),$$

with $v_i^{(\mu)}, v_i^{(n-\mu)} \in \{0, \dots, \frac{n}{(n, l_i)} - 1\}$. Taking the last equation mode_i gives that:

$$v_i^{(\mu)} + v_i^{(n-\mu)} \equiv 0 \text{ mode}_i.$$

Therefore:

$$v_i^{(\mu)} \equiv 0 \text{ mode}_i \Leftrightarrow v_i^{(n-\mu)} \equiv 0 \text{ mode}_i \Leftrightarrow v_i^{(\mu)} = 0 \Leftrightarrow v_i^{(n-\mu)} = 0.$$

Otherwise $v_i^{(\mu)} + v_i^{(n-\mu)} = e_i$. We conclude that:

$$m_i^{(\mu)} + m_i^{(n-\mu)} = \begin{cases} l_i - 1 & \text{for } i \in I^{(\mu)} \\ l_i & \text{otherwise,} \end{cases}$$

hence:

$$(7) \quad g_\mu(x) \cdot g_{n-\mu}(x) = \frac{\prod_{i=1}^r (x - \rho_i)^{l_i}}{\prod_{i \in I^{(\mu)}} (x - \rho_i)} = \frac{f(x)}{\prod_{i \in I^{(\mu)}} (x - \rho_i)}.$$

Going back to the calculation of $d\left(\frac{y^\mu}{x^\nu g_\mu(x)}\right)$ in eq. (6) using eq. (7) we have:

$$\begin{aligned}d\left(\frac{y^\mu}{x^\nu g_\mu(x)}\right) &= \frac{\prod_{i \in I^{(\mu)}} (x - \rho_i) \left(x \sum_{j \in I^{(\mu)}} \frac{v_j^{(\mu)}(n, l_j)}{x - \rho_j} - \nu n\right)}{n x^{\nu+1}} \omega_{n-\mu, 1} \\ &= \frac{\psi_{\mu, \nu}(x)}{n x^{\nu+1}} \omega_{n-\mu, 1} \\ &= \frac{\psi_{\mu, \nu}^{\leq \nu+1}(x)}{n x^{\nu+1}} \omega_{n-\mu, 1} - \frac{\psi_{\mu, \nu}^{> \nu+1}(x)}{n x^{\nu+1}} \omega_{n-\mu, 1},\end{aligned}$$

where of course the first summand is in Ω_0 , and the second summand is in Ω_∞ , as we wanted. $\square$

3. ARTIN-SCHREIER EXTENSIONS

3.1. Computations. We consider the curve given as an Artin-Schreier extension of $k(x)$, where k is again an algebraically closed field of characteristic $p \geq 3$, given by an equation of the form

$$X : y^p - y = r(x) = \frac{f(x)}{\prod_{i=1}^r (x - \rho_i)^{l_i}}.$$

Without loss of generality we can assume that $f(x)$ is prime to the denominator of $r(x)$, that $(l_i, p) = 1$ for all $i \in \{1, \dots, r\}$, that $x \nmid f(x)$ (i.e. 0 is not a root of $f(x)$), and that the points above ∞ are neither poles nor roots for y (which means that $\deg f(x) = \sum_{i=1}^r l_i =: l$). This means that ∞ is not in the branch locus of the cover $X \rightarrow \mathbb{P}^1$ and the points above it are $P_{\infty, 1}, \dots, P_{\infty, p}$. For any given

Artin-Schreier curve, we can use Möbius transformations of the base curve $\mathbb{P}^1$ to make it satisfy the above conditions. The points of X lying above each $x = \rho_i$ are P_i with ramification index p .

If $0 \in \mathbb{P}^1$ is in the branch locus, then $g_0 := 1$ and $e_0 := p$. If 0 is not in the branch locus, then $g_0 := p$ and $e_0 := 1$.

We collect several known facts for the divisors of functions and differentials, see [1, sec. 3]

$$(x) = e_0 \sum_{j=1}^{g_0} P_{0,j} - \sum_{m=1}^p P_{\infty,m}$$

$$(y) = (y)_0 - \sum_{i=1}^r l_i P_i = (f(x))_0 - \sum_{i=1}^r l_i P_i,$$

where in the root divisor $(y)_0 = (f(x))_0$ of y , none of the P_i 's appears. Moreover

$$\left(\prod_{i=1}^r (x - \rho_i)^{l_i} \right) = p \sum_{i=1}^r l_i P_i - l \sum_{m=1}^p P_{\infty,m}$$

$$(y^{p-1} - 1) = A - (p-1) \sum_{i=1}^r l_i P_i$$

and A is the root part of the divisor of $y^{p-1} - 1$ where the roots of $f(x)$ appear.

We will also need the divisor of the differential dx :

$$(dx) = \sum_{i=1}^r (p-1)(l_i + 1) P_i - 2 \sum_{m=1}^p P_{\infty,m},$$

hence for μ a positive integer we get

$$(y^\mu dx) = \mu(y)_0 + \sum_{i=1}^r ((p-1-\mu)l_i + (p-1)) P_i - 2 \sum_{m=1}^p P_{\infty,m}.$$

For each μ , $\mu \in \{1, 2, \dots, p-1\}$, we define $m_i^{(\mu)}$ and $v_i^{(\mu)}$ as the quotient and the residue, respectively, of the following Euclidean division

$$(p-1-\mu)l_i + (p-1) = m_i^{(\mu)}p + v_i^{(\mu)}, \text{ with } 0 \leq v_i^{(\mu)} \leq p-1.$$

We set

$$g_\mu(x) = \prod_{i=1}^r (x - \rho_i)^{m_i^{(\mu)}} \text{ for } \mu \in \{1, 2, \dots, p-1\}$$

and we have that

$$(g_\mu(x)) = p \sum_{i=1}^r m_i^{(\mu)} P_i - t^{(\mu)} \sum_{m=1}^p P_{\infty,m},$$

where

$$t^{(\mu)} = \sum_{i=1}^r m_i^{(\mu)}.$$

Now, we compute

$$\left(\frac{y^\mu}{g_\mu(x)} dx \right) = \mu(y)_0 + \sum_{i=1}^r v_i^{(\mu)} P_i + (t^{(\mu)} - 2) \sum_{m=1}^p P_{\infty,m}.$$

Finally we will need

$$\begin{aligned}
(g_{p-\mu}(x)y^{\mu-1}) &= p \sum_{i=1}^r m_i^{(p-\mu)} P_i - t^{(p-\mu)} \sum_{m=1}^p P_{\infty,m} \\
(8) \quad &+ (\mu-1)(f(x))_0 - \sum_{i=1}^r (\mu-1)l_i P_i \\
&= \sum_{i=1}^r (pm_i^{(p-\mu)} - (\mu-1)l_i) P_i + (\mu-1)(f(x))_0 - t^{(p-\mu)} \sum_{m=1}^p P_{\infty,m}
\end{aligned}$$

where $pm_i^{(p-\mu)} - (\mu-1)l_i = p-1 - v_i^{(p-\mu)} \geq 0$.

3.2. A k -vector space basis for $H^1(X, \mathcal{O}_X)$. As in the Kummer case, we denote by $\pi : X \rightarrow \mathbb{P}^1$ a covering map from X to $\mathbb{P}^1$. For every point $a \in \mathbb{P}^1$ we define $U_a := X \setminus \pi^{-1}(a)$. Observe that by the calculation (8) the elements

$$\frac{g_{p-\mu}(x)y^{\mu-1}}{x^\nu} \text{ for } 1 \leq \mu \leq p-1, \text{ s.t. } t^{(p-\mu)} \geq 2, 1 \leq \nu \leq t^{(p-\mu)} - 1$$

have poles only above zero and infinity, i.e. they are in $\mathcal{O}_X(U_0 \cap U_\infty)$. By identifying the first sheaf cohomology group of $\mathcal{O}_X$ with the corresponding first de Rham cohomology group, then we get that

$$H^1(X, \mathcal{O}_X) = \frac{\mathcal{O}_X(U_0 \cap U_\infty)}{\{f_0 - f_\infty : f_0 \in \mathcal{O}_X(U_0), f_\infty \in \mathcal{O}_X(U_\infty)\}}.$$

From Serre duality, this $H^1(X, \mathcal{O}_X)$ seen as a k -vector space, is dual to the k -vector space $H^0(X, \Omega_X)$. From the calculations of the previous paragraph and from [1] we can see that a basis of $H^0(X, \Omega_X)$ is given by the differentials

$$\omega_{\mu,\nu} := x^{\nu-1} \frac{y^\mu}{g_\mu(x)} dx, 1 \leq \mu \leq p-1, \text{ s.t. } t^{(\mu)} \geq 2, 1 \leq \nu \leq t^{(\mu)} - 1.$$

Also, the following linear equivalence classes

$$h_{\mu,\nu} := \left[\frac{g_{p-\mu}(x)y^{\mu-1}}{x^\nu} \right] \text{ with } 1 \leq \mu \leq p-1, \text{ s.t. } t^{(p-\mu)} \geq 2, 1 \leq \nu \leq t^{(p-\mu)} - 1.$$

are well defined, linearly independent elements of $H^1(X, \mathcal{O}_X)$. Since they are in bijective correspondence with the elements $\omega_{\mu,\nu}$, we know that they form a basis (the k -vector space dimension of $H^0(X, \Omega_X)$ is equal to the k -vector space dimension of $H^1(X, \mathcal{O}_X)$).

Remark. Note that the condition $t^{(\mu)} \geq 2$ means that $\mu < p-1$, because for $\mu = p-1$ we have $m_i^{(\mu)} = 0$ for all i . For the same reason the condition $t^{(p-\mu)} \geq 2$ means that $\mu > 1$.

As in the section of the Kummer case, using the map

$$\begin{aligned}
\langle \cdot, \cdot \rangle : H^0(X, \Omega_X) \times H^1(X, \mathcal{O}_X) &\longrightarrow k \\
\langle \omega, [f] \rangle &\longmapsto \sum_{P \in \pi^{-1}(\infty)} \text{Res}_P(f\omega)
\end{aligned}$$

we will show that:

Proposition 4. *The bases of $H^0(X, \Omega_X)$ and $H^1(X, \mathcal{O}_X)$ that we have written above are dual to each other.*

Proof. We calculate the pairing between elements of the above given bases:

$$\begin{aligned}
\langle \omega_{\mu_1, \nu_1}, h_{\mu_2, \nu_2} \rangle &= \sum_{P \in \pi^{-1}(\infty)} \text{Res}_P \left(\frac{g_{p-\mu_2}(x) y^{\mu_2-1} x^{\nu_1-1} y^{\mu_1}}{x^{\nu_2} g_{\mu_1}(x)} dx \right) \\
&= \sum_{P \in \pi^{-1}(\infty)} \text{Res}_P \left(\frac{y^{\mu_1+\mu_2-1} g_{p-\mu_2}(x) x^{\nu_1-\nu_2}}{g_{\mu_1}(x)} \frac{dx}{x} \right) \\
&= \text{Res}_\infty \left(\text{Tr}_{F/E} (y^{\mu_1+\mu_2-1}) \frac{g_{p-\mu_2}(x) x^{\nu_1-\nu_2}}{g_{\mu_1}(x)} \frac{dx}{x} \right)
\end{aligned}$$

We have to calculate the trace $y^{\mu_1+\mu_2-1}$ for the various values of μ_1 and μ_2 . Notice that $1 \leq \mu_1 + \mu_2 - 1 \leq 2(p-1) - 1$, so we will write the trace of y^k for $k \in \{1, \dots, 2(p-1) - 1\}$:

$$\begin{aligned}
\text{Tr}_{F/E}(y^k) &= \sum_{i=0}^{p-1} (y+i)^k = y^k + \sum_{i=1}^{p-1} (y+i)^k \\
&= y^k + \sum_{i=1}^{p-1} \sum_{j=0}^k \binom{k}{j} y^{k-j} i^j \\
&= y^k + \sum_{j=0}^k \binom{k}{j} y^{k-j} \sum_{i=1}^{p-1} i^j \\
&= y^k + y^k \left(\sum_{i=1}^{p-1} i^0 \right) + \sum_{j=1}^k \binom{k}{j} y^{k-j} \sum_{i=1}^{p-1} i^j \\
&= \sum_{j=1}^k \binom{k}{j} y^{k-j} \sum_{i=1}^{p-1} i^j
\end{aligned}$$

since $\sum_{i=1}^{p-1} i^0 = p-1 = -1$ in characteristic p , and this cancels out with the first y^k .

Let's focus on the inner sum of the above expression: For $j \in \{1, \dots, p-2, p, \dots, 2(p-1) - 1\}$, we have:

$$\sum_{i=1}^{p-1} i^j = \sum_{t \in \mathbb{F}_p^\times} t^j = \sum_{l=0}^{p-2} (g^l)^j$$

where g is a generator of $\mathbb{F}_p^\times$. Since $j \in \{1, \dots, p-2, p, \dots, 2(p-1) - 1\}$ we have $g^j \neq 1$, and thus

$$\sum_{i=1}^{p-1} i^j = \sum_{l=0}^{p-2} (g^j)^l = \frac{1 - (g^j)^{p-1}}{1 - g^j} = \frac{1 - (g^{p-1})^j}{1 - g^j} = 0$$

where the last sum is the sum of terms of a geometric progression, and of course $g^{p-1} = 1$ for $g \in \mathbb{F}_p^\times$.

On the other hand for $j = p-1$:

$$\sum_{i=1}^{p-1} i^j = \sum_{t \in \mathbb{F}_p^\times} t^{p-1} = p-1 = -1 \text{ in characteristic } p.$$

However for $k \geq p$, the binomial coefficient $\binom{k}{p-1}$ is divisible by p , hence it is zero in characteristic p .

From all the above we conclude that:

$$\mathrm{Tr}_{F/E}(y^k) = \begin{cases} -1 & \text{for } k = p-1 \\ 0 & \text{for } k \in \{1, \dots, p-2, p, \dots, 2(p-1)-1\}. \end{cases}$$

Returning to the calculation of the pairing, we see that the result is non-zero only for $\mu_1 + \mu_2 - 1 = p-1$, i.e. for $\mu_1 = p - \mu_2$. In that case:

$$\begin{aligned} \langle \omega_{p-\mu_2, \nu_1}, h_{\mu_2, \nu_2} \rangle &= \mathrm{Res}_\infty \left((-1) \frac{g_{p-\mu_2}(x) x^{\nu_1-\nu_2}}{g_{p-\mu_2}(x)} \frac{dx}{x} \right) \\ &= \mathrm{Res}_\infty \left(\left(\frac{1}{x} \right)^{\nu_2-\nu_1} \frac{d\left(\frac{1}{x}\right)}{\left(\frac{1}{x}\right)} \right). \end{aligned}$$

From the properties of the residues, the last residue from above is non-zero if and only if $\nu_1 = \nu_2$, in which case it is equal to 1, and this completes our proof. $\square$

3.3. A k -vector space basis for $H_{dR}^1(X/k)$. Similarly to the case of a Kummer extension we can compute $H_{dR}^1(X/k)$ as follows:

$$\frac{\{(\omega_0, \omega_\infty, f_{0\infty}) \in \Omega_X(U_0) \times \Omega_X(U_\infty) \times \mathcal{O}_X(U_0 \cap U_\infty) \mid df_{0\infty} = \omega_0|_{U_0 \cap U_\infty} - \omega_\infty|_{U_0 \cap U_\infty}\}}{\{(df_0, df_\infty, f_0|_{U_0 \cap U_\infty} - f_\infty|_{U_0 \cap U_\infty}) \mid f_0 \in \mathcal{O}_X(U_0), f_\infty \in \mathcal{O}_X(U_\infty)\}}.$$

Again, we have canonical maps

$$\begin{aligned} i : H^0(X, \Omega_X) &\rightarrow H_{dR}^1(X/k) \\ \omega &\mapsto [(\omega|_{U_0}, \omega|_{U_\infty}, 0)] \end{aligned}$$

and

$$\begin{aligned} p : H_{dR}^1(X/k) &\rightarrow H^1(X, \mathcal{O}_X) \\ [(\omega_0, \omega_\infty, f_{0\infty})] &\mapsto [f_{0\infty}]. \end{aligned}$$

Same as in the previous section, from Proposition 4.1.2 of Tait's dissertation ([16]) we get that the following short sequence is exact:

$$0 \rightarrow H^0(X, \Omega_X) \xrightarrow{i} H_{dR}^1(X/k) \xrightarrow{p} H^1(X, \mathcal{O}_X) \rightarrow 0$$

hence the basis of the k -vector space $H_{dR}^1(X/k)$ can be calculated from the bases of $H^0(X, \Omega_X)$ and $H^1(X, \mathcal{O}_X)$.

Theorem 5. *Let*

$$\varphi_{\mu, \nu}(x) := x g'_{p-\mu}(x) g_{\mu-1}(x) - \nu g_{p-\mu}(x) g_{\mu-1}(x) \in k[x]$$

for $1 \leq \mu \leq p-1$ s.t. $t^{(p-\mu)} \geq 2$ and $1 \leq \nu \leq t^{(p-\mu)} - 1$. Also let

$$\psi(x) := f(x) \sum_{i=1}^r l_i \left(\prod_{j=1, j \neq i}^r (x - \rho_j) \right) - f'(x) \prod_{i=1}^r (x - \rho_i) \in k[x].$$

Let

$$\omega_\mu := \frac{(\mu-1) g_{p-\mu}(x) y^{\mu-2}}{\prod_{i=1}^r (x - \rho_i)^{l_i+1}} dx$$

for $1 \leq \mu \leq p-1$ s.t. $t^{(p-\mu)} \geq 2$. Let us denote

$$\tilde{h}_{\mu, \nu} := \frac{g_{p-\mu}(x) y^{\mu-1}}{x^\nu}.$$

The elements

$$a_{\mu,\nu} := \left[\left(\frac{\varphi_{\mu,\nu}^{<\nu+1}(x)}{x^{\nu+1}} \omega_{\mu-1,1} + \frac{\psi^{<\nu}(x)}{x^\nu} \omega_\mu, \frac{\varphi_{\mu,\nu}^{>\nu+1}(x)}{x^{\nu+1}} \omega_{\mu-1,1} + \frac{\psi^{>\nu}(x)}{x^\nu} \omega_\mu, \tilde{h}_{\mu,\nu} \right) \right]$$

for $1 \leq \mu \leq p-1$ s.t. $t^{(p-\mu)} \geq 2$ and $1 \leq \nu \leq t^{(p-\mu)} - 1$

together with the elements

$$\delta_{\mu,\nu} := \left[\left(x^{\nu-1} \frac{y^\mu}{g_\mu(x)} dx, x^{\nu-1} \frac{y^\mu}{g_\mu(x)} dx, 0 \right) \right]$$

for $1 \leq \mu \leq p-1$ s.t. $t^{(\mu)} \geq 2$ and $1 \leq \nu \leq t^{(\mu)} - 1$

form a basis for $H_{dR}^1(X/k)$ as a k -vector space.

Proof. Similarly to the corresponding theorem about Kummer extensions, we just need to show that the $a_{\mu,\nu}$'s are well-defined elements of $H_{dR}^1(X/k)$.

First we write down the differential of $\tilde{h}_{\mu,\nu}$:

$$\begin{aligned} d \left(\frac{g_{p-\mu}(x)y^{\mu-1}}{x^\nu} \right) &= \frac{(g'_{p-\mu}(x)y^{\mu-1} - \nu g_{p-\mu}(x)y^{\mu-1}x^{-1})dx + (\mu-1)g_{p-\mu}(x)y^{\mu-2}dy}{x^\nu} \\ &= \frac{xg'_{p-\mu}(x)g_{\mu-1}(x) - \nu g_{p-\mu}(x)g_{\mu-1}(x)}{x^{\nu+1}} \left(\frac{y^{\mu-1}}{g_{\mu-1}(x)} dx \right) \\ &\quad + \frac{(\mu-1)g_{p-\mu}(x)y^{\mu-2}dy}{x^\nu} \\ &= \frac{\varphi_{\mu,\nu}(x)}{x^{\nu+1}} \omega_{\mu-1,1} + \frac{(\mu-1)g_{p-\mu}(x)y^{\mu-2}}{x^\nu} dy. \end{aligned}$$

For the differential of y we can use the defining equation of our curve, since we are in characteristic p :

$$\begin{aligned} d(y^p - y) &= d \left(\frac{f(x)}{\prod_{i=1}^r (x - \rho_i)^{l_i}} \right) \\ \Rightarrow -dy &= \frac{f'(x) - f(x) \sum_{i=1}^r \frac{l_i}{x - \rho_i}}{\prod_{i=1}^r (x - \rho_i)^{l_i}} \\ \Rightarrow dy &= \frac{f(x) \sum_{i=1}^r l_i \left(\prod_{j=1, j \neq i}^r (x - \rho_j) \right) - f'(x) \prod_{i=1}^r (x - \rho_i)}{\prod_{i=1}^r (x - \rho_i)^{l_i+1}} \\ \Rightarrow dy &= \frac{\psi(x)}{\prod_{i=1}^r (x - \rho_i)^{l_i+1}}. \end{aligned}$$

Thus, we end up with:

$$d \left(\frac{g_{p-\mu}(x)y^{\mu-1}}{x^\nu} \right) = \frac{\varphi_{\mu,\nu}(x)}{x^{\nu+1}} \omega_{\mu-1,1} + \frac{\psi(x)}{x^\nu} \omega_\mu.$$

We will show that for all $\mu \in \{1, \dots, p-1\}$ s.t. $t^{(p-\mu)} \geq 2$, ω_μ is either holomorphic or has pole of order one at each point above ∞ . Then it's easily seen that the $a_{\mu,\nu}$'s are well-defined, so the proof will be over.

Notice that ω_μ can have poles only at the points above ∞ (we'll show that when that happens the order of the pole is one) or at the P_i 's (we'll show that this never happens), since $f(x)$ never appears in the denominator - remember that $\mu \geq 2$ from the remark we made after writing the basis of $H^1(X, \mathcal{O}_X)$ above.

Firstly, let $P_{\infty, m}$ be any point above ∞ . Then:

$$\begin{aligned} v_{P_{\infty, m}}(\omega_\mu) &= v_{P_{\infty, m}}(g_{p-\mu}(x)) + v_{P_{\infty, m}}(dx) + v_{P_{\infty, m}}\left(\prod_{i=1}^r (x - \rho_i)^{l_i+1}\right) \\ &= -t^{(p-\mu)} - 2 + \sum_{i=1}^r (l_i + 1) \\ &= -2 + \sum_{i=1}^r (l_i - m_i^{(p-\mu)} + 1), \end{aligned}$$

where for all $i \in \{1, \dots, r\}$

$$(p-1)(l_i + 1) - (p-\mu)l_i = pm_i^{(p-\mu)} + v_i^{(p-\mu)},$$

which leads to

$$(p-1)(l_i - m_i^{(p-\mu)} + 1) = (p-\mu)l_i + m_i^{(p-\mu)} + v_i^{(p-\mu)}.$$

The right hand side of the last equation is strictly positive, so the same is true for the left hand side, and more specifically

$$l_i - m_i^{(p-\mu)} + 1 > 0, \text{ for all } i \in \{1, \dots, r\}.$$

Therefore if $r > 1$, then $\sum_{i=1}^r (l_i - m_i^{(p-\mu)} + 1) \geq 2$ and ω_μ has no pole at any point above ∞ . If $r = 1$, then $l_1 - m_1^{(p-\mu)} + 1 \geq 1$ and ω_μ is either holomorphic (if $l_1 > m_1^{(p-\mu)}$) or has a pole at each $P_{\infty, m}$ of order one (if $l_1 = m_1^{(p-\mu)}$; for example $p = 7, \mu = 6, l_1 = 2$ give $m_1^{(1)} = 2 = l_1$).

Now let's check whether ω_μ has pole at any P_i :

$$\begin{aligned} v_{P_i}(\omega_\mu) &= v_{P_i}(g_{p-\mu}(x)) + v_{P_i}(y^{\mu-2}) + v_{P_i}(dx) - v_{P_i}\left(\prod_{i=1}^r (x - \rho_i)^{l_i+1}\right) \\ &= pm_i^{(p-\mu)} - (\mu-2)l_i + (p-1)(l_i + 1) - p(l_i + 1) \\ &= (\mu-1)l_i + p-1 - v_i^{(p-\mu)} - (\mu-2)l_i - l_i - 1 \\ &= p-2 - v_i^{(p-\mu)}. \end{aligned}$$

If $v_i^{(p-\mu)} = p-1$, then $(\mu-1)l_i \equiv 0 \pmod{p}$ with $(p, l_i) = 1$ by assumption, so $\mu = 1$, but that's not allowed from the remark we made after writing the basis for $H^1(X, \mathcal{O}_X)$. Hence $v_i^{(p-\mu)} \leq p-2$, which implies that $v_{P_i}(\omega_\mu) \geq 0$, as we wanted. $\square$

REFERENCES

- [1] Helmut Boreck. Zur Theorie der Weierstrasspunkte. *Math. Nachr.*, 19:29–63, 1958.
- [2] Hara Charalambous, Kostas Karagiannis, and Aristides Kontogeorgis. The relative canonical ideal of the Artin-Schreier-Kummer-Witt family of curves. *Annales de l'institut Fourier*, 73(3):1085–1113, 2023.
- [3] Jędrzej Garnek. Equivariant splitting of the Hodge–de Rham exact sequence. *Math. Z.*, 300(2):1917–1938, 2022.
- [4] Jędrzej Garnek. p -group Galois covers of curves in characteristic p . *Trans. Amer. Math. Soc.*, 376(8):5857–5897, 2023.
- [5] Jędrzej Garnek and Aristides Kontogeorgis. The de Rham cohomology of covers with a cyclic p -Sylow subgroup. *J. Algebra*, 687:151–178, 2026.

- [6] Jędrzej Garnek. p -group Galois covers of curves in characteristic p . II. *Doc. Math.*, 30(2):347–377, 2025.
- [7] Helmut Hasse. Theorie der relativ-zyklischen algebraischen Funktionenkörper, insbesondere bei endlichem Konstantenkörper. *J. Reine Angew. Math.*, 172:37–54, 1934.
- [8] Sotiris Karanikolopoulos and Aristides Kontogeorgis. Integral representations of cyclic groups acting on relative holomorphic differentials of deformations of curves with automorphisms. *Proc. Amer. Math. Soc.*, 142(7):2369–2383, 2014.
- [9] Bernhard Köck and Joseph Tait. On the de-Rham cohomology of hyperelliptic curves. *Res. Number Theory*, 4(2):Paper No. 19, 17, 2018.
- [10] Aristides Kontogeorgis. The group of automorphisms of cyclic extensions of rational function fields. *J. Algebra*, 216(2):665–706, 1999.
- [11] Aristides I. Kontogeorgis. The group of automorphisms of the function fields of the curve $x^n + y^m + 1 = 0$. *J. Number Theory*, 72(1):110–136, 1998.
- [12] T. Sekiguchi, F. Oort, and N. Suwa. On the deformation of Artin-Schreier to Kummer. *Ann. Sci. École Norm. Sup. (4)*, 22(3):345–375, 1989.
- [13] Tsutomu Sekiguchi and Noriyuki Suwa. Théories de Kummer-Artin-Schreier-Witt. *C. R. Acad. Sci. Paris Sér. I Math.*, 319(2):105–110, 1994.
- [14] Tsutomu Sekiguchi and Noriyuki Suwa. Théorie de Kummer-Artin-Schreier et applications. *J. Théor. Nombres Bordeaux*, 7(1):177–189, 1995. Les Dix-huitièmes Journées Arithmétiques (Bordeaux, 1993).
- [15] Jean-Pierre Serre. *Algebraic groups and class fields*, volume 117 of *Graduate Texts in Mathematics*. Springer-Verlag, New York, 1988. Translated from the French.
- [16] Joseph J. Tait. *Group actions on differentials of curves and cohomology bases of hyperelliptic curves*. PhD thesis, University of Southampton, Southampton, United Kingdom, November 2014. Faculty of Social and Human Sciences, Mathematical Sciences.

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